

Theory of Demand, Supply and Market Equilibrium

HSS102: Concepts in Humanities and Social Sciences (Market)

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Different types of goods

B. External factors (Economic + Non-economic)

B.1. Economic Factors

1. Price of related goods

Substitute goods: A product that can be used in **place** of another product

$$\bullet P_{\text{coffee}} \uparrow \rightarrow Q_{\text{dd for coffee}} \downarrow \rightarrow DD_{\text{tea}} \uparrow$$

Complementary goods: A product that is used in **conjunction** with another product

$$P_{\text{pen}} \uparrow \rightarrow Q_{\text{dd for pen}} \downarrow \rightarrow DD_{\text{ink}} \downarrow$$

B. External factors (Economic + Non-economic)

B.1. Economic Factors

2. Consumer's income

Normal goods: a product that increases in demand with an increase in income (eg., car, books, clothes)

Inferior: a product that decreases in demand with an increase in income (eg., public transport, low-grade rice)

Factors Influencing Demand

B.2. Non-economic factor

1. Tastes and Preferences

- » Consumers must first desire or have tastes and preferences for a good.
- » Socioeconomic variables such as age, gender, race, marital status, and level of education are often good proxies for an individual's tastes and preferences for a particular good, because tastes and preferences may vary by these groupings and products are often targeted at one or more of these groups.
- » Economic theory may also suggest that one or more of these socioeconomic variables influences the demand for a particular good or service.
- » **Tastes and preferences encompass all the individualistic variables that influence a person's willingness to purchase a good.**

2. Taxation

3. Advertisement

4. Expectation about future prices

5. Festive seasons

6. Climate change

Changes in Quantity Demanded and Change in Demand

- Changes in Quantity Demanded
- Changes in Demand

Exceptional Cases with Price Change (Demand Curve)

- Giffen goods (Robert Giffen)
 - (upward sloping demand curve and with few/no substitution) [eg., essential commodities]
- Veblen goods (Thorstein Veblen)
 - (upward sloping demand curve) [eg., expensive commodities]

Individual Demand and Market Demand

- Each consumer has his or her own **individual demand function** for different products.
- Managers are usually more interested in the **market demand function**, which shows the quantity demanded of the good or service by *all* consumers in the market at any given price.
- The market demand function is influenced by the prices of related goods, as well as by the tastes and preferences, income, and future expectations of all consumers in the market. It can also change because more consumers enter the market.
- **Horizontal summation of individual demand curves:**
The process of deriving a **market demand curve** by adding the quantity demanded by each individual at every price to determine the market demand at every price.

Supply

- The functional relationship between the price of a good or service and the quantity supplied by producers in a given time period, *all else held constant*.
- The supply curve shows the quantity of a good a firm is willing to produce at each price. Normally a firm is willing to produce more as the price increases, which is why **the supply curve slopes upward**.

- Quantity supplied
 - the amount of a good that firms want to sell at a given price, holding constant other factors that influence firms' supply decisions, such as costs and government actions
- Supply curve
 - the quantity supplied at each possible price, holding constant the other factors that influence firms' supply decisions

An example

- Suppose that (keeping everything else constant) the supply function for the commodity is given by $Q_s = 1000P$, where Q_s stands for the market quantity supplied of the commodity per time period and P for the price of the commodity.
 - (a) Derive the market supply schedule for this commodity
 - (b) Draw the market supply curve for this commodity.

Market Supply Survey

- The market supply curve shows the quantity of a good all firms in the market are willing to supply at each price.
- The market supply curve is normally upward sloping, both because each firm is willing to supply more of the good at a higher price and because higher prices entice new firms to produce.

Factors influencing Supply Function

- ***State of Technology***

- The state of technology, or the body of knowledge about how to combine the inputs of production, affects what output producers will supply because technology influences how the good or service is actually produced, which, in turn, affects the costs of production.

- ***Input Prices***

- Input prices are the prices of all the inputs or factors of production—labour, capital, land, and raw materials—used to produce the given product. These input prices affect the costs of production and, therefore, the prices at which producers are willing to supply different amounts of output.

- ***Prices of Goods Related in Production***

- The prices of other goods related in production can also affect the supply of a particular good. Two goods are substitutes in production if the same inputs can be used to produce either of the goods, such as land for different agricultural crops.

- ***Future Expectations***

- Future expectations can play a role on the supply side of the market as well. If producers expect prices to increase in the future, they may supply less output now than without those expectations.

- ***Number of Producers***

- The number of producers influences the total supply of a product at any given price. The number of producers may increase because of perceived profitability in a given industry or because of changes in laws or regulations such as trade barriers.

- **Government rules and regulations**

- **Externalities**

- Cost of other factors which has no direct relations.

Changes in Quantity Supplied and Changes in Supply

Equilibrium price

- » The price that actually exists in the market or toward which the market is moving where the quantity demanded by consumers equals the quantity supplied by producers.

Equilibrium quantity

- » The quantity of a good, determined by the equilibrium price, where the amount of output that consumers demand is equal to the amount that producers want to supply.

An example

1. $Q_d=500-10P$ & $Q_s=50+5P$, find out equilibrium quantity and price.
2. $Q_d=160-40P$ & $Q_s=50+15P$, find out equilibrium quantity and price.

Shocking to the equilibrium

- Change in demand
- Change in supply